Results Analysis and Reporting

1. Summary of Performance Metrics

Steady-State Run (Main Test)

- **Average Response Time:** ~188 ms
- **P90 Response Time:** ~210 ms
- **P95 Response Time:** ~255 ms
- **Throughput:** ~1 request/second
- **Error Rate:** ~0.9%

Smoke Test

- **Average Response Time:** ~200–220 ms
- **P90 Response Time:** ~226 ms
- **P95 Response Time:** ~344 ms
- **Throughput:** very low (single user validation)
- **Error Rate:** 0%

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2. Comparison Between Runs

Smoke Run

- Executed with 1 user
- Purpose: validate correctness of the script
- Result: 100% success rate
- No errors observed
- Response times slightly higher due to small sample size

Assertions

Assertion	Status
Global: percentage of successful events is greater than 95.0	OK
Global: 95th percentile of response time is less than 1500.0	OK

Expand all groups | Collapse all groups

Requests	Executions					Response Time (ms)							
	Total	OK	KO	% KO	Cnt/s	Min	50th pct	75th pct	95th pct	99th pct	Max	Mean	Std Dev
All Requests	4	4	0	0	0.5	155	226	344	344	344	344	235	68
Login Request	1	1	0	0	0.12	344	344	344	344	344	344	344	0
Get Current User	2	2	0	0	0.25	216	226	226	226	226	226	221	5
Refresh Session	1	1	0	0	0.12	155	155	155	155	155	155	155	0

Steady-State Run

- Executed with 10 concurrent users
- Includes ramp-up, steady duration, and ramp-down
- Stable response times across all endpoints
- Minor failure rate (~3.5%) observed only in login

Assertions

Assertion	Status
Global: percentage of successful events is greater than 85.0	OK
Global: 95th percentile of response time is less than 2500.0	OK
Login Request: percentage of failed events is less than 15.0	OK
Get Current User: percentage of failed events is less than 5.0	OK
Refresh Session: percentage of failed events is less than 5.0	OK

Expand all groups | Collapse all groups

Requests	Executions					Response Time (ms)							
	Total	OK	KO	% KO	Cnt/s	Min	50th pct	75th pct	95th pct	99th pct	Max	Mean	Std Dev
All Requests	2228	2208	20	0.9	0.99	141	162	210	255	511	1032	188	69
Login Request	572	552	20	3.5	0.25	141	166	193	266	700	1032	189	88
Get Current User	1104	1104	0	0	0.49	142	168	237	261	498	865	192	64
Refresh Session	552	552	0	0	0.24	143	151	193	253	409	698	178	55

Errors

Error	Count	Percentage
status.find.is(200), found 400	20	100%

Key Differences

Aspect	Smoke	Steady
Users	1	10
Purpose	Validation	Performance
Errors	0%	~0.9% overall
Login Stability	Perfect	Slightly sensitive
Load Behavior	Not representative	Realistic

Explanation

- Smoke test does not represent real load → used only for validation
- Steady test introduces concurrency → reveals real system behavior
- Failures appear only under concurrent usage → expected in performance testing

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3. Observed Facts

- Login endpoint has ~3.5% failure rate under load
- Get Current User and Refresh endpoints have 0% failure rate
- Overall system error rate remains low (~0.9%)
- Response times remain stable under load (P95 < 300 ms)
- No HTTP 500 errors observed
- System maintains consistent throughput (~1 req/sec)

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4. Bottleneck Hypothesis

Based on observed behavior:

- The login endpoint is the most sensitive component
- Failures are likely caused by:
    - concurrent authentication requests
    - session/token overwriting
    - shared-user conflicts (initial runs)

Conclusion:

- The issue is **not a system crash**
- It is related to **authentication handling under concurrency**

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5. Test Limitations

- Target system is a **public API (DummyJSON)**
- Environment is shared and not controlled
- Limited number of valid user accounts
- No access to backend metrics (CPU, DB, etc.)
- Results may vary between executions

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6. Conclusion

The system demonstrates strong performance under moderate concurrent load, maintaining low response times and minimal error rates.

The primary risk lies in the authentication endpoint, which shows sensitivity under concurrent usage, especially when test data is limited.

Key Priorities

- Improve authentication scalability
- Ensure proper session/token isolation
- Increase available test data for realistic simulation

Next Actions

- Perform higher-scale stress testing
- Monitor authentication service under peak conditions
- Introduce better user data distribution strategies